Memorize for final exam (thermal)

Tuesday, April 21, 2026 11:39 AM

$z = \sum e^{-E(s)/kT}$ $C_V = \left(\frac{\partial \langle E \rangle}{\partial T}\right)_{V,N}$

harmonic oscillator: $E_n = (n+\frac{1}{2})\hbar\nu$

velocity space

$v_{rms} = \sqrt{\frac{3kT}{m}}$

$P(v) = \boxed{4\pi v^2}\, e^{-E(s)/kT}$

or in higher dim : $P(s) \sim v^{d-1} e^{-E(s)/kT}$

normalize $\int_0^\infty P(v)\,dv = 1$

most probable speed : $\frac{dP(v)}{dv} = 0$

average speed : $v_{avg} = \int_0^\infty v P(v)\,dv$

Quantum / classical limit

quantum volume where classical breaks down :

$$\underbrace{L = \sqrt{\frac{h^2}{2\pi m kT}}}_{\text{de Broglie wavelength}} \Rightarrow V_q = L^3$$

dim = 3

energy = kT

$V = z_1 V_q$ ← BOLTZMANN LIMIT

box vol.

how many quantum cells in box (in quantum, $z_1 \sim N$)

one thermal quantum cell volume (λ_T^3)

for N noninteracting particles:

$z_N = \frac{1}{N!} z_1^N$

$= \frac{1}{N!}\left(\frac{V}{V_q}\right)^N$ ← BOLTZMANN LIMIT

If there are internal dof,

$z_N = \frac{1}{N!}\left(\frac{V z_{int}}{V_q}\right)^N$

particle in 3D box: $E_n = \frac{n^2h^2}{8mL^2}$

Deriving properties w/ partition function

Total Energy : $U = -\frac{1}{z}\frac{\partial z}{\partial \beta}$

Helmholtz free energy : $F = -kT\ln z$

Heat capacity : $C_V = \frac{dU}{dT}$

Pressure : $P = -\left(\frac{dF}{dV}\right)_{T,N}$

Entropy : $S = -\left(\frac{dF}{dT}\right)_{V,N}$

Chemical potential : $\mu = \left(\frac{dF}{dN}\right)_{T,V}$

PV charts

$\Delta U = W + Q$ ON system

$W_{on} = -\int_{V_1}^{V_2} P\,dV$

ADIABATIC : $PV^\gamma = \text{const.}$

$TV^{\gamma-1} = \text{const.}$

$\Delta U_{on} = \frac{f}{2}Nk(T_2 - T_1)$

$= \frac{f}{2}(P_2V_2 - P_1V_1)$

Efficiency / COP

Engine : $e = \frac{W}{Q_h}$ ← $W = Q_h - Q_c$

$= 1 - \frac{Q_c}{Q_h}$

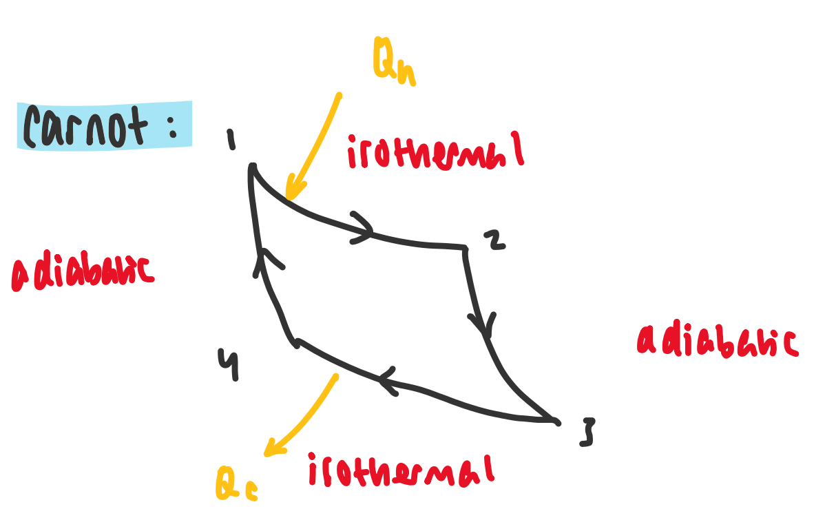
max : $e \leq 1 - \frac{T_c}{T_h}$ (carnot)

Refrigerator: $COP = \frac{Q_c}{W} = \frac{Q_c}{Q_h - Q_c} = \frac{1}{\frac{Q_h}{Q_c} - 1}$

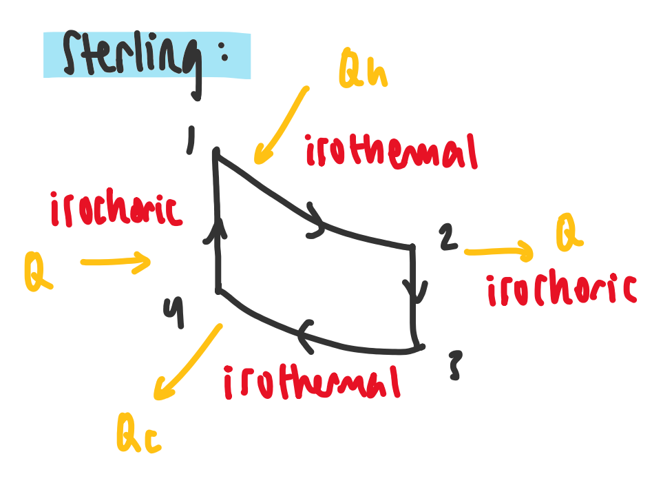
max COP: $COP \leq \underbrace{\frac{T_c}{T_h - T_c}}$

smaller ΔT = larger COP

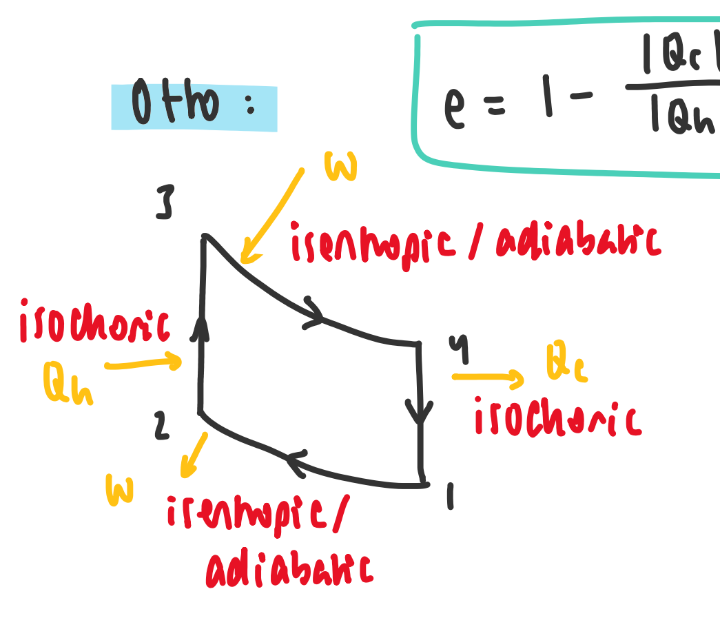
Carnot :



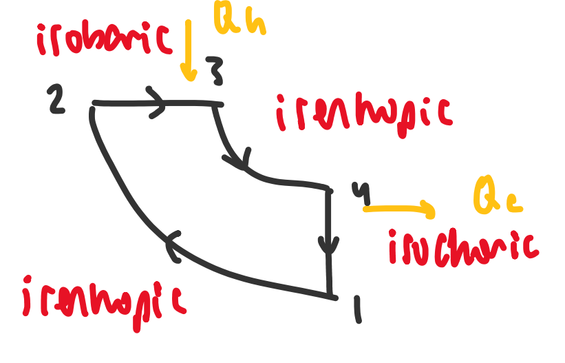
Sterling :



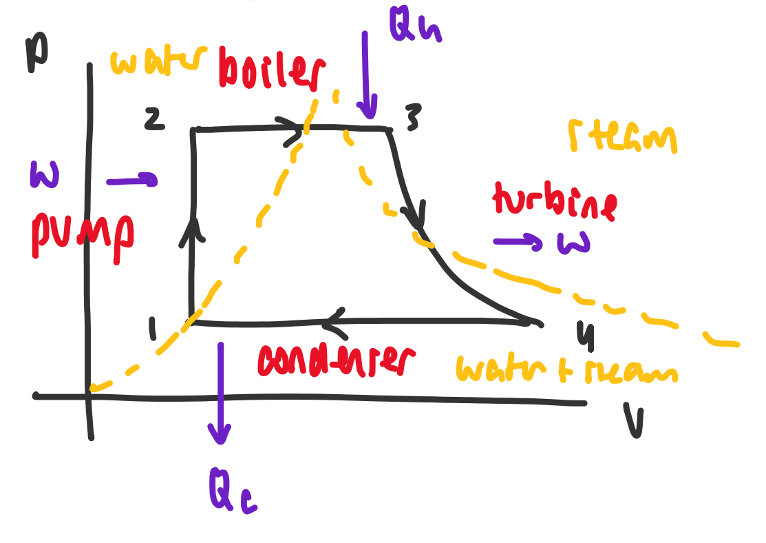
Otto : $\boxed{e = 1 - \frac{|Q_c|}{|Q_h|}} = 1 - \left(\frac{V_2}{V_1}\right)^{\gamma-1}$



Diesel : $e = 1 - \left(\frac{V_2}{V_1}\right)^{\gamma-1}\left[\frac{\alpha^\gamma - 1}{\gamma(\alpha - 1)}\right]$



Steam engine: Rankine



$e = \frac{W}{Q_{in}} = \frac{(H_3 - H_4) + (H_1 - H_2)}{H_3 - H_2}$ given

$= 1 - \frac{H_1 - H_4}{H_3 - H_2}$

$\boxed{H_2 \approx H_1}$

$3 \to 4$: isentropic

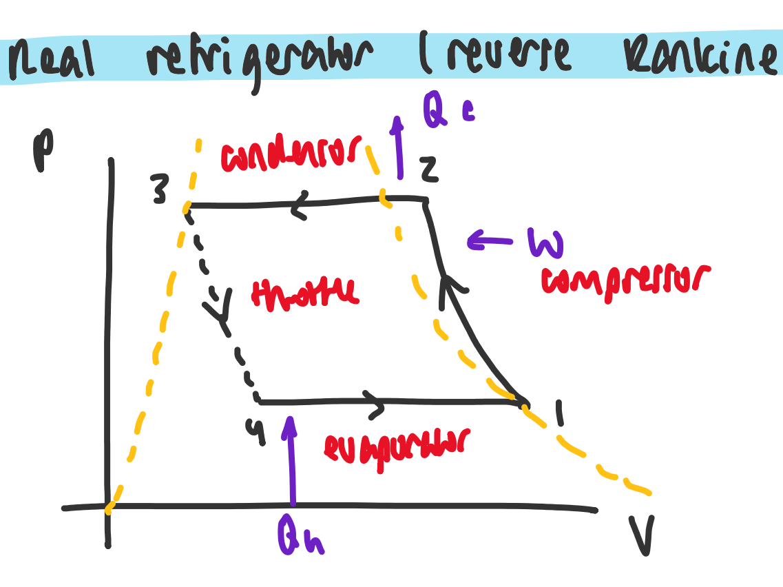
① find H_1 on table (water). $H_1 \sim H_2$

② find S_3 on table. $S_3 \sim S_4$

③ get ratio of water and steam using S_4 (water + steam)

④ use this ratio to get H_4

Real refrigerator (reverse Rankine)

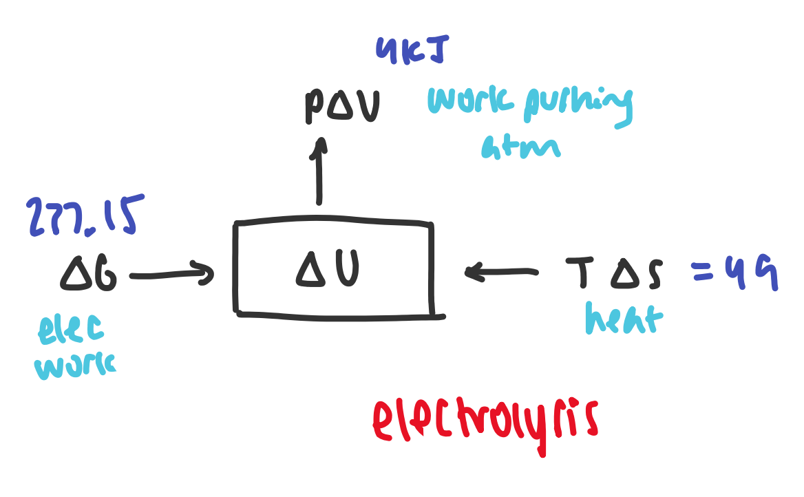


$COP = \frac{Q_c}{W}$

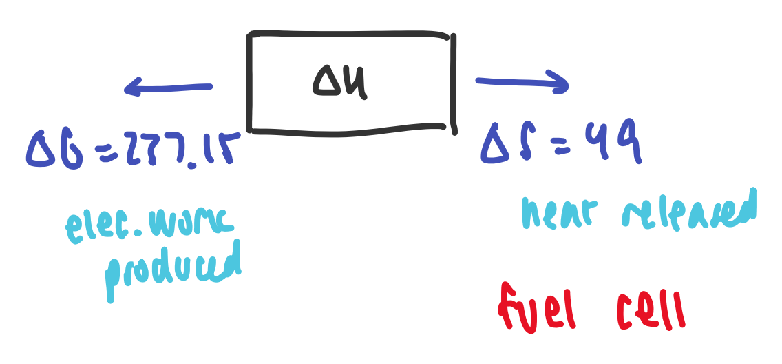
$= \frac{Q_c}{Q_h - Q_c}$

$= \frac{1}{\frac{Q_h}{Q_c} - 1}$

Gibbs free energy



electrolysis



fuel cell

$W_{electron} = \frac{\Delta G}{N_{electrons} \times N_A}$ [eV] $\Rightarrow V = \frac{W_{electron}}{e}$ [V]

$Pb + PbO_2 + 4H^+ + 2SO_4^{2-} \rightarrow 2PbSO_4 + 2H_2O$

added and cancelled

voltage : solution : $2SO_4^{2-} + 2H^+ \rightarrow 2HSO_4^-$

cathode : $PbO_2 + HSO_4^- + 3H^+ + 2e^- \rightarrow PbSO_4 + 2H_2O$

anode : $Pb + HSO_4^- \rightarrow PbSO_4 + H^+ + 2e^-$

$W_{electron} = \frac{390\text{kJ}}{(2)(6.02 \times 10^{23})}$

$= 2.02\text{eV}$

$\frac{dP}{dT} = \frac{L}{T\Delta V}$ ← pressure to inc/dec. melting point by 1°C

ΔV: $V_{gas} - V_{liquid}$

osmotic pressure : $P = i(P_2 - P_1) = i\frac{NkT}{V} = i\frac{nRT}{V}$

$\frac{N}{V}$: solute molecules / unit volume

i: degree of dissociation

↳ non-dissociating: $i = 1$

↳ NaCl : $i = 2$

Le chatelier: $N_2 + 3H_2 \leftrightarrow 2NH_3$

$\frac{P^2_{NH_3}}{P_{N_2}P^3_{H_2}} = K = -e^{-\frac{G^\circ}{kT}}$

higher pressure favours side with more "space"

① $CO_2(g) + H_2O(l) \leftrightarrow H_2CO_3(aq)$

$\frac{m_{H_2CO_3}}{P/P_0} = e^{-\frac{\Delta G^\circ}{RT}}$

for dissolving

② $H_2CO_3 \leftrightarrow H^+ + HCO_3^-$

dissociation : assume equal

$\frac{m_{H^+} m_{HCO_3^-}}{m_{H_2CO_3}} = K = e^{-\frac{\Delta G^\circ}{RT}}$

for dissociation

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partition function

$z = \sum e^{-E(s)/kT}$

$P(s) = \frac{1}{z}e^{-E(s)/kT}$

Rotational for diatomic: $z_{rot} \sim \frac{kT}{\varepsilon}$

$\bar{E}_{rot} = kT$

symmetry (for O_2, N_2) : $z_{rot} \sim \frac{kT}{2\varepsilon}$

$\bar{E}_{rot} = kT$ still